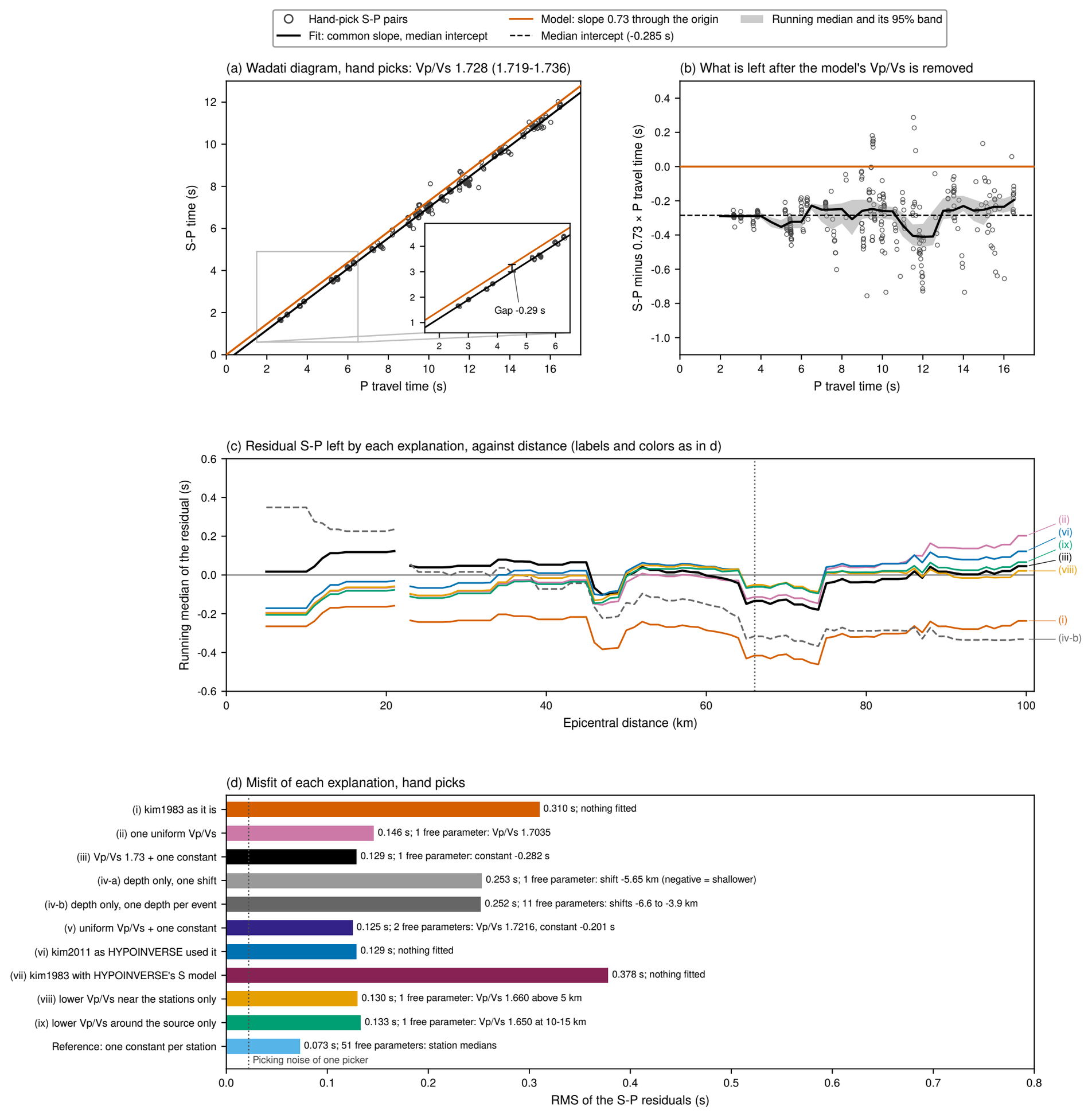
Hand-pick S-P pairs
Fit: common slope, median intercept
Model: slope 0.73 through the origin
Median intercept (-0.285 s)
Running median and its 95% band
(a) Wadati diagram, hand picks: Vp/Vs 1.728 (1.719-1.736)
S-P time (s)
P travel time (s)
Gap -0.29 s
(b) What is left after the model's Vp/Vs is removed
S-P minus 0.73 × P travel time (s)
P travel time (s)
(c) Residual S-P left by each explanation, against distance (labels and colors as in d)
Running median of the residual (s)
Epicentral distance (km)
(ii)
(vi)
(ix)
(iii)
(viii)
(i)
(iv-b)
(d) Misfit of each explanation, hand picks
(i) kim1983 as it is
0.310 s; nothing fitted
(ii) one uniform Vp/Vs
0.146 s; 1 free parameter: Vp/Vs 1.7035
(iii) Vp/Vs 1.73 + one constant
0.129 s; 1 free parameter: constant -0.282 s
(iv-a) depth only, one shift
0.253 s; 1 free parameter: shift -5.65 km (negative = shallower)
(iv-b) depth only, one depth per event
0.252 s; 11 free parameters: shifts -6.6 to -3.9 km
(v) uniform Vp/Vs + one constant
0.125 s; 2 free parameters: Vp/Vs 1.7216, constant -0.201 s
(vi) kim2011 as HYPOINVERSE used it
0.129 s; nothing fitted
(vii) kim1983 with HYPOINVERSE's S model
0.378 s; nothing fitted
(viii) lower Vp/Vs near the stations only
0.130 s; 1 free parameter: Vp/Vs 1.660 above 5 km
(ix) lower Vp/Vs around the source only
0.133 s; 1 free parameter: Vp/Vs 1.650 at 10-15 km
Reference: one constant per station
0.073 s; 51 free parameters: station medians
Picking noise of one picker
RMS of the S-P residuals (s)